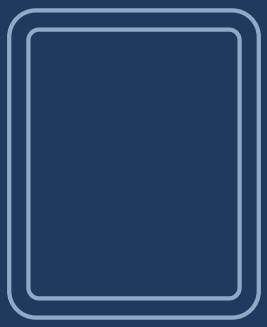
KPS PUBLIC SCHOOLS • STUDENT TECH BENCH PROGRAM

Lenovo Chromebook Duet EDU G2



Solder-Free Repair Guide

Every repair a student bench can safely do, without soldering.
Lenovo Chromebook Duet EDU G2: detachable tablet with keyboard folio.

In this guide

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Tablet design: most internal parts are glued.
Part numbers: look up by serial on pcsupport.lenovo.com.

1 Read first

This guide covers every Lenovo Chromebook Duet EDU G2 repair the Tech Bench can do with hand tools: swapping a part, reseating a cable, or fixing software. Anything that needs a soldering iron, hot air, or board-level work is in Section 6 and goes to the lead tech or the manufacturer.

How this guide was built. Lenovo hasn't published a public service manual for the Duet EDU G2. Like most tablets, the screen and battery are held with adhesive, so this guide keeps student work to the keyboard folio, charging port and software, and marks the glued-in repairs as lead-tech jobs. Steps follow standard glued-tablet practice; record what you find in the notes boxes.

Most repairs are lead-tech jobs. Opening a glued tablet risks cracking the screen and puncturing the battery. Students do Level 1 work; the lead tech does Level 3.

Bench levels

Level	Who may do it	Typical repairs
LEVEL 1	Any trained student	Software fixes, keyboard folio, pogo pins, USB-C port cleaning
LEVEL 2	Student with supervisor sign-off on this repair	Speaker checks
LEVEL 3	Lead tech or adult supervisor only	Screen, battery, camera (all need the screen off)

Before you touch any device

1. ☐ Check the asset tag and look the device up in the district inventory. Note any open ticket.
2. ☐ Ask the lead tech whether the unit is still under Lenovo warranty or an accidental-damage plan. A warranty claim may cost less than a part, and opening a unit can affect warranty coverage.
3. ☐ Photograph all four sides of the device before opening it.
4. ☐ Record the symptom in the user's own words on the ticket.
5. ☐ Warn the user that a reset erases local files (Downloads). Cloud files in Drive are safe.

Safety rules

- **Power off, unplug, remove any SD card and cables,** then disconnect the battery before any other work inside.
- **Lithium batteries:** never puncture, bend, or pry against the battery. A swollen, hot, hissing, or smoking battery is a Level 3 stop. Move it to the metal battery bin or sand and get an adult.
- **ESD:** wear a grounded wrist strap on an ESD mat. Touch boards only by the edges.
- **Ribbon cables:** unlock every ZIF latch before sliding a ribbon out. A locked ribbon tears.
- **Glass:** wear safety glasses when removing a cracked screen.
- **Before power-on:** check for loose screws inside. Shake gently and listen. A loose screw can short the board.

Tools and supplies

Item	Use	Notes
Plastic picks and suction cup	Screen removal	Lead tech only
Heat pad or hair dryer	Softening adhesive	Low heat, keep it moving
Replacement adhesive strips	Resealing the screen	Order with the screen
Plastic spudger, opening picks, plastic tweezers	Clips, latches, ribbons	Only plastic touches connectors
ESD mat + wrist strap	Static protection	
Magnetic screw mat	Keeping screws in order	Screws are often different lengths
90%+ isopropyl alcohol + soft brush	Cleaning contacts, spill residue	
Kapton tape	Re-taping cables and connectors	
Recovery USB or SD card, 8 GB+	ChromeOS recovery	Chromebook Recovery Utility
Known-good USB-C charger	Testing power and charging	
Safety glasses	Cracked-screen removal	
Fire-safe battery bin or sand bucket	Damaged batteries	Required at every bench

2 Know the machine

Specs that matter for repair

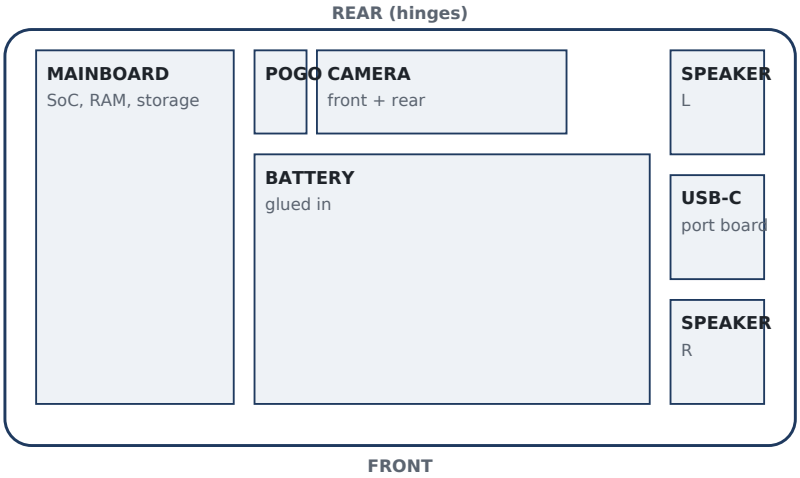
Item	Duet EDU G2	Why it matters
Form	10.95" detachable tablet with magnetic keyboard folio	Folio connects by pogo pins
Processor / memory / storage	MediaTek; soldered	Board level
Screen	Glass, touch and LCD in one glued assembly	Pen support
Battery	Glued in	Lead tech
Ports	USB-C (charging, display, data)	
Recovery keys	Volume Up + Volume Down + Power	See S3

How it comes apart

The screen is glued to the frame. Under it: the battery (glued), system board, cameras and speakers. The keyboard folio and kickstand cover attach with magnets and pogo pins.

Inside the tablet

With the screen lifted off you would see the parts below. It's a rough map; correct it if the lead tech opens one.



ILLUSTRATIVE — block positions are approximate. Confirm on your first teardown.

Duet EDU G2 parts list

Lenovo part numbers depend on the build. Look up each part with the serial number on PartsLookup (pcsupport.lenovo.com).

Part	Part number	Repair
Keyboard folio / kickstand cover	by serial	R1
Display assembly + adhesive	by serial	R3
Battery	by serial	R4
Cameras	by serial	R5
Speakers	by serial	R6

3 Triage: symptom → repair

Start with the cheapest, least invasive fix. Work through "try first" before ordering parts.

Symptom	Try first	Then	Part likely needed
Won't turn on	Known-good charger 30 min → S1	R2	Battery or board
Not charging / loose charger	Other charger → port cleaning	R2 → Sec. 6	USB-C port
Battery drains fast	S2 battery test	R4	Battery
Keyboard dead or intermittent	Detach, clean pins	R1	Folio or pogo connector
Keys type wrong symbols	S4	S4	None
Cracked screen / dead touch	External display test	R3	Display assembly
Camera not detected	Privacy; S1; S3	R5	Camera
No sound	Headphone test	R6	Speakers
Wi-Fi drops	S3, then forget and rejoin network	S3	Board (Sec. 6)
Swollen battery / screen lifting	STOP — Level 3	R4	Battery

4 Software repairs

No tools needed. Always do S1 before opening a device.

S1 Hard reset (EC reset) LEVEL 1

Fixes	Won't power on, won't charge, frozen, black screen, unresponsive keyboard or touchpad
Time	2 minutes
Tools	Known-good USB-C charger
Data loss	None

A hard reset restarts the embedded controller (power, keyboard and charging chip). It erases nothing.

Procedure

1. ☐ Unplug the charger and any USB devices. Remove any SD card.
2. ☐ Hold Power for 10 seconds so the device is fully off.
3. ☐ Hold **Volume Up + Power** for about 10 seconds, then release.
4. ☐ If nothing happens: plug in a known-good charger, wait 30 minutes, and repeat with the charger connected. Try the other USB-C port too.

Verify before closing the ticket

- Device reaches the login screen.
- Charging light comes on with the charger connected.

Tip. After a battery has been disconnected, plug in a charger before deciding the device is dead. Many Chromebooks won't wake on the power button alone.

S2 Built-in hardware diagnostics LEVEL 1

Fixes	Confirms the fault before ordering a part: battery, CPU, memory, keyboard, touchpad, Wi-Fi
Time	5–15 minutes
Tools	None
Data loss	None

Procedure

1. ☐ Sign in with a bench test account if you can.
2. ☐ Open the Launcher, search **Diagnostics**, and open it.
3. ☐ **System:** read battery health and cycle count, then run the battery tests.
4. ☐ Run the CPU and memory tests.
5. ☐ **Connectivity:** confirm Wi-Fi is detected and run the tests.
6. ☐ **Input:** open the keyboard tester and press every key; test the touchpad and touchscreen.
7. ☐ Screenshot any failed result (Ctrl + Show windows) and attach it to the ticket.

Verify before closing the ticket

- Results recorded on the ticket.
- Parts ordered only for a failed test or a fault you saw yourself.

S3 Powerwash and USB recovery

LEVEL 1

Fixes	Slow, crashing, corrupted OS, "ChromeOS is missing or damaged", Wi-Fi missing after an update
Time	Powerwash 10 min; recovery 20–40 min
Tools	Recovery USB or SD card (8 GB+)
Data loss	**Yes.** Erases all local files (Downloads).

District-managed devices. Enrolled devices should force re-enrollment on their first Wi-Fi connection after a wipe. Confirm with the lead tech that forced re-enrollment is on and the bench Wi-Fi reaches enrollment. Never enter Developer Mode on a district device.

Procedure

1. ☐ **Check for updates first:** Settings > About ChromeOS > Check for updates.
2. ☐ **Powerwash:** sign out, press Ctrl + Alt + Shift + R, choose Restart, then Powerwash. (May be blocked by policy; use recovery instead.)
3. ☐ **Make recovery media:** on a working computer, open the Chromebook Recovery Utility, choose the **Lenovo Chromebook Duet EDU G2** from its list (or search the model), insert an 8 GB+ drive, and click Create now.
4. ☐ Hold **Volume Up + Volume Down + Power** for about 10 seconds, then release when the recovery screen appears.
5. ☐ Insert the recovery media and follow the prompts.
6. ☐ After it reboots, connect to Wi-Fi and confirm it re-enrolls to the KPS domain.

Verify before closing the ticket

- Boots to the enrollment or KPS sign-in screen.
- Shows as enrolled in the Admin console.
- Recovery can't write to storage: storage has failed. That's a system board repair or warranty claim.

S4 Keyboard layout and international keyboard fix

LEVEL 1

Fixes	Keys type the wrong symbols (@ and " swapped, accents appear, Z and Y swapped); a replacement keyboard with a different layout
Time	5 minutes
Tools	None
Data loss	None

ChromeOS can have several keyboard layouts installed. Students often switch layouts by accident with **Ctrl + Shift + Space**, or have **US International** on, which turns ' and " into accent keys. The physical keyboard is fine.

Procedure

1. ☐ Look at the bottom-right shelf. A layout code (US, INTL, UK, ES...) means more than one layout is installed.
2. ☐ Open **Settings > Device > Keyboard and inputs > Input** (on older versions: Settings > Advanced > Languages and inputs > Inputs).
3. ☐ Set **English (US)** as the active input. Remove any layouts the student doesn't need, especially US International, unless a teacher asked for it.
4. ☐ Test in a text box: Shift + 2 should type @ and Shift + ' should type ".
5. ☐ If a **replacement keyboard** has a different physical layout (UK keyboard on a US unit, for example), order the correct US part. Changing the software layout won't fix printed key labels.
6. ☐ If the wrong layout keeps coming back for everyone who signs in, it's set by district policy. Tell the lead tech; don't change it on the device.

Verify before closing the ticket

- Every key types its printed character in the S2 keyboard tester and in a text box.
- Only the needed layouts remain installed.

5 Hardware repairs

Only the first two repairs are student-level. Everything behind the glass is a lead-tech job.

Golden rules. Photograph every layer before removing it. Lay screws on the mat in the same pattern as the device. Unlock every connector before pulling a cable. Never force a part: if it won't move, find the screw or clip you missed. Test before the last screws go in.

R1 Keyboard folio replacement and pogo-pin cleaning — LEVEL 1

Fixes	Keyboard dead or intermittent, broken folio, kickstand cover damaged
Part	Keyboard folio / kickstand cover for the Duet EDU G2 (look up by serial)
Time	5–10 minutes
Tools	Isopropyl alcohol, cotton swab, soft brush

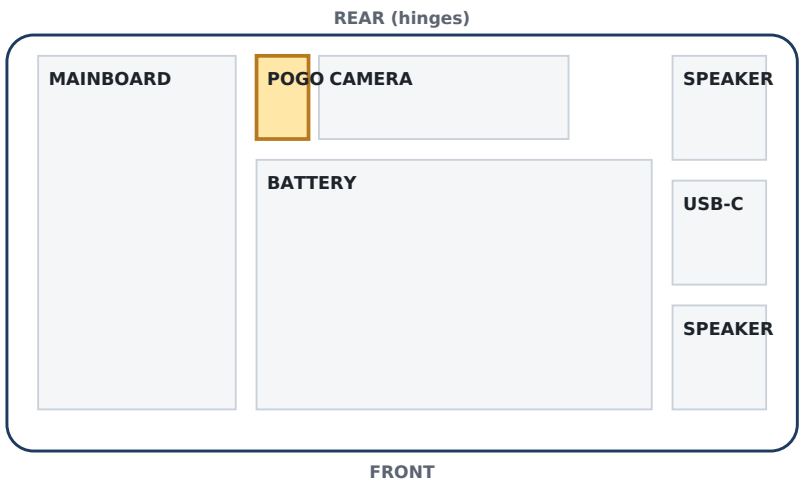


Illustration: tablet from the back. Positions illustrative.

The detachable keyboard has no internal repairs: it's replaced as a unit.

Procedure

1. ☐ Detach the keyboard from the tablet.
2. ☐ Inspect the pogo pins on the tablet edge and the contacts on the keyboard: look for dirt, corrosion or a bent pin.
3. ☐ Clean both with a swab lightly dampened in isopropyl alcohol. Let them dry 2 minutes.
4. ☐ Reattach. The magnets should pull the keyboard into line and the screen should show the keyboard is connected.
5. ☐ Still dead after cleaning: try a known-good keyboard. If that works, replace the folio. If it doesn't, the tablet's pogo connector is faulty (Section 6).
6. ☐ Kickstand cover: peel the old cover off its magnets and press the new one on, lining up the camera cut-out.

Verify before closing the ticket

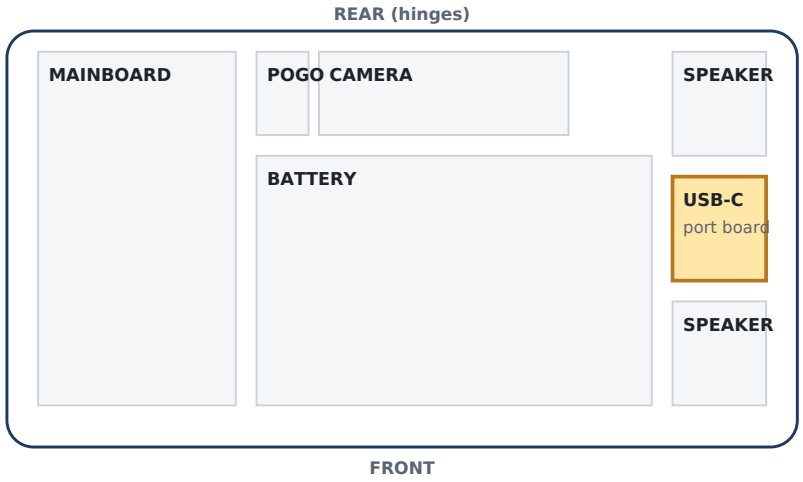
- Every key passes the S2 keyboard test.

- Touchpad on the folio moves and clicks.
- Keyboard reconnects after detaching 5 times.

R2 USB-C charging check and port cleaning

LEVEL 1

Fixes	Not charging, charges only at an angle, charger falls out
Part	None, or the port board (lead tech)
Time	10 minutes
Tools	Wooden toothpick, flashlight, known-good charger



1. Look for lint at the back of the port.
2. Wooden toothpick only.
3. Wiggle test after cleaning.

Illustration: tablet from the back. Positions illustrative.

Procedure

1. ☐ Try a known-good charger and cable in the USB-C port.
2. ☐ Shine a light into the port. Lint packed at the back is common.
3. ☐ Power off. Gently scrape lint out with a wooden toothpick (never metal).
4. ☐ Retest. Wiggle the cable: if charging cuts in and out, the port is worn: Section 6.

Verify before closing the ticket

- Charges with the cable straight and at a slight angle.
- Battery percentage rises over 10 minutes.

R3 Screen (display assembly) replacement — lead tech

LEVEL 3

Fixes	Cracked glass, dead touch, lines or black screen with an external display working
Part	Display assembly (glass, touch and LCD in one part; look up by serial)
Time	60–90 minutes
Tools	Plastic picks, plastic spudger, heat pad or hair dryer on low, suction cup, new adhesive, safety glasses

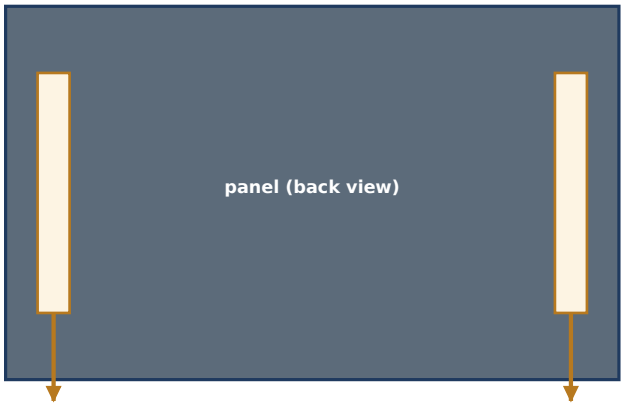


Illustration: stretch-release panel tape. Tape position illustrative.

1. Bezel off first. The panel has no screws.
2. Find the stretch-release tape tabs under the left and right edges.
3. Pull each tab slowly, almost flat to the screen, with tweezers. It may take several pulls.
4. Never pry the panel off over unreleased tape; it cracks.
5. Fit new tape strips on reassembly.

Lead tech only. The screen is glued on. A slip with the pick can puncture the battery or cut the display cable. Do this repair only with the lead tech, and only with a replacement screen in hand.

Check first: does an external USB-C display show a picture? If yes, the fault is the screen; if no, it's the board (Section 6).

Procedure

1. ☐ Power off. Detach the keyboard. Tape across cracked glass.
2. ☐ Warm the screen edges evenly (about 80 °C on a heat pad, or a hair dryer on low, moving) for 2–3 minutes.
3. ☐ Lift one edge with a suction cup and slide a thin pick under the glass. Keep the pick shallow (under 3 mm) so it doesn't cut the display cable.
4. ☐ Work the pick around all four sides, re-warming as needed. Don't twist or flex the glass.
5. ☐ Lift the screen like a book. Disconnect the battery first, then the display connector.
6. ☐ Clean all old adhesive off the frame. Fit new adhesive strips.
7. ☐ Connect the new screen and the battery, and test picture and touch **before** sealing.
8. ☐ Press the screen down evenly and keep it clamped or weighted for an hour.

Verify before closing the ticket

- Picture full, no lines; touch works in every corner.
- No gaps around the edge; no creaking when pressed.
- Rotation and the cameras work.

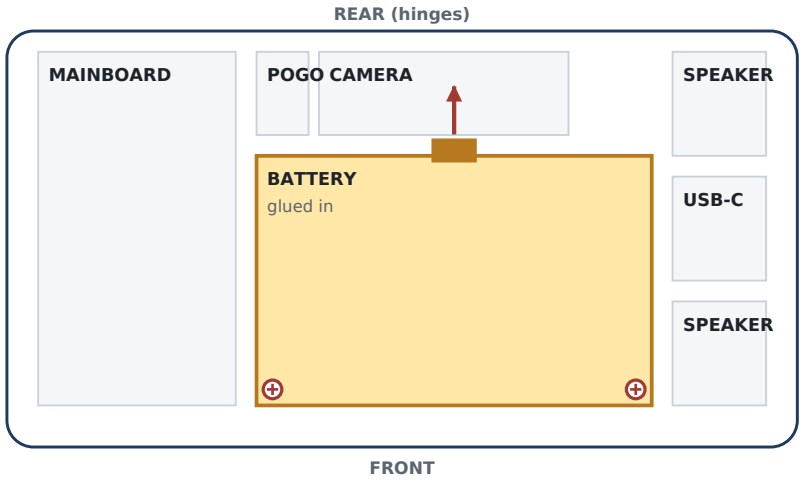
First-teardown notes — fill in the first time you do this repair

Where the display connector sits and how deep the pick could safely go. _____

R4 Battery replacement — lead tech

LEVEL 3

Fixes	Won't hold charge, low health in S2, swollen battery
Part	Battery (look up by serial)
Time	60–90 minutes (includes screen removal)
Tools	Plastic picks, plastic spudger, isopropyl alcohol, plastic card, ESD strap



1. Unplug the battery cable from the mainboard first (red arrow).
2. Disconnect the battery before anything else; plastic tools only.
3. Remove the battery screws (count them).
4. Lift out flat. Never bend or pry against the cell.

Illustration: base interior, battery highlighted. Screw positions illustrative.

Swollen or punctured battery. Stop. Don't press or pry. A punctured lithium cell can catch fire. The lead tech moves it to the fire-safe bin.

Procedure

1. ☐ Remove the screen (previous repair).
2. ☐ Disconnect the battery connector from the board.
3. ☐ The battery is glued. Drip a little isopropyl alcohol under one edge and wait a minute.
4. ☐ Slide a plastic card under the battery and lift slowly and flat. **Never use metal; never bend the cell.**
5. ☐ Fit the new battery, connect it, test, then refit the screen with new adhesive.

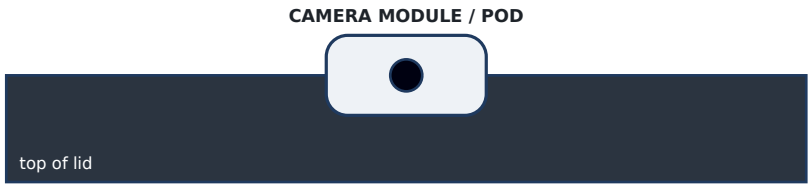
Verify before closing the ticket

- Battery percentage shows and charging light comes on.
- S2 battery health near 100% on a new battery.

R5 Camera replacement — lead tech

LEVEL 3

Fixes	Camera not detected or blurry with the lens clean
Part	Front or rear camera module (look up by serial)
Time	With screen removal, 60–90 minutes
Tools	Plastic picks, plastic spudger



1. Bezel off first (it covers the camera's front).
2. Unplug the small camera cable with tweezers.
3. Lift the module by its board edges, never the lens.
4. Test in the Camera app before closing up.

Illustration: camera at the top-center of the lid. Cap shapes simplified.

Procedure

1. ☐ Check first: privacy settings, S1 and a powerwash (S3).
2. ☐ Remove the screen (screen repair) and disconnect the battery.
3. ☐ Disconnect the camera connector, lift the module out, fit the new one and reconnect.
4. ☐ Reconnect the battery and test before resealing the screen.

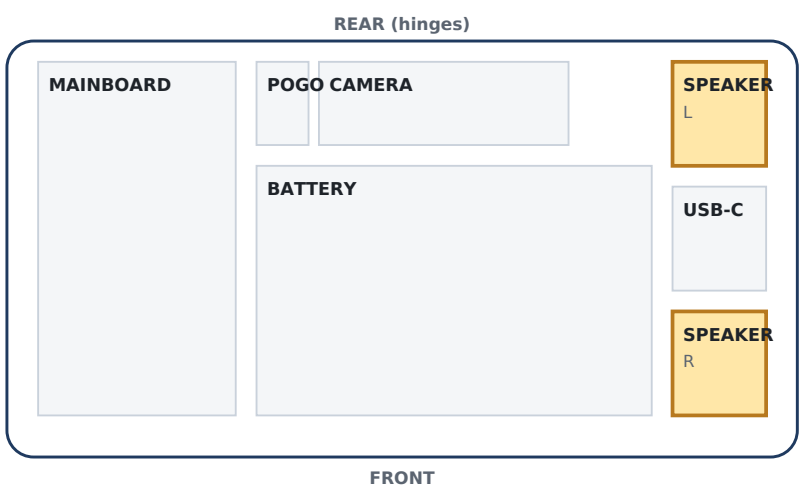
Verify before closing the ticket

- Camera app shows a clear image on both cameras.
- Works in a Google Meet test call.

R6 Speaker check

LEVEL 2

Fixes	No sound or crackling with headphones working
Part	Speaker modules (look up by serial) — requires screen removal
Time	5 minutes to check
Tools	Soft brush



1. Speakers sit at both short ends.
2. Replacement needs the screen off.
3. Lead tech only.

Illustration: tablet from the back. Positions illustrative.

Procedure

1. ☐ Check volume, mute and the output device in quick settings.
2. ☐ Brush lint out of the speaker grilles gently.
3. ☐ Test with a USB-C headphone adapter: sound there but not from speakers = hardware.
4. ☐ Speaker replacement needs the screen off: treat it like the camera repair, with the lead tech.

Verify before closing the ticket

- Clear sound from both sides at full volume.

6 Out of scope: escalate

These need soldering or board-level work, or parts that aren't sold. Tag the device with the ticket and send it to the lead tech.

Fault	Why	Route
USB-C port worn or loose	Port board behind the glued screen	Lead tech or Lenovo warranty
Pogo connector on the tablet dead	Behind the glued screen	Lenovo warranty
Wi-Fi hardware failure	Built into the system board	Lenovo warranty
Storage or RAM failure	Soldered	Lenovo warranty
Burn marks, melted plastic, burnt smell	Fire hazard	Lead tech immediately
Any repair needing a soldering iron or hot air	Program policy	Lenovo warranty or lead tech

Lenovo support

- **Lenovo support:** pcsupport.lenovo.com (enter the serial number)
- **Parts lookup:** PartsLookup on pcsupport.lenovo.com

7 Sources

Checked October 2026. The procedures are the guide author's adaptation of these sources for a student bench; they are not manufacturer-issued instructions.

- Lenovo PC Support (pcsupport.lenovo.com) — specifications and parts lookup by serial.
- Standard glued-tablet service practice (heat, plastic tools, adhesive replacement) as published in iFixit's tablet repair guides.